100 Programs

to Build Your Problem-Solving

For absolute beginners — any language (Python / Java / C++)

Before you touch LeetCode or DSA, finish these 100.
Start from Program 1 and go in order — basics first, then moderate, then challenge.
Wherever you see **n**, it is taken as user input.

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How to use this sheet

1. Pick **one** language and stick to it for all 100 — don't keep switching.

2. Solve strictly in order. The problems are arranged from easiest to hardest for a reason.

3. Do NOT look up the solution first. Struggle for 10–15 minutes on your own — that struggle is where problem-solving is actually built.

4. Type the code yourself every single time. No copy-paste.

5. After each program works, ask: "Can I do it with fewer lines or a cleaner logic?"

6. Wherever **n** appears, read it from the user as input.

7. Tick off each problem as you finish. Aim for 3–5 per day; you'll be done in about a month.

Rule of thumb: Finish these 100 first. Then go anywhere — DSA, LeetCode, projects.

Level 1 — Warm-up (Output, variables, basic input)

1. Write a program to print "Hello, World!" on the screen.
2. Write a program to read two numbers and print their sum.
3. Write a program to read two numbers and print their sum, difference, product and quotient.
4. Write a program to read the radius of a circle and print its area and circumference.
5. Write a program to read the length and breadth of a rectangle and print its area and perimeter.
6. Write a program to swap two numbers using a third variable.
7. Write a program to swap two numbers without using a third variable.
8. Write a program to read a temperature in Celsius and convert it to Fahrenheit.
9. Write a program to read the marks of 5 subjects and print the total and average.
10. Write a program to read seconds and convert them into hours, minutes and seconds.

Level 2 — Conditions (if / else)

11. Write a program to read a number and check whether it is even or odd.
12. Write a program to read a number and check whether it is positive, negative or zero.
13. Write a program to read three numbers and find the largest among them.
14. Write a program to read three numbers and find the smallest among them.
15. Write a program to read a year and check whether it is a leap year or not.
16. Write a program to read a character and check whether it is a vowel or a consonant.
17. Write a program to read a character and check whether it is an alphabet, digit or special symbol.
18. Write a program to read the marks of a student and print the grade (A/B/C/D/Fail).
19. Write a program to read a number and check whether it is divisible by both 3 and 5.
20. Write a program to read the age of a person and check whether they are eligible to vote.

Level 3 — Loops on N (n is user input)

21. Write a program to display all the natural numbers from 1 to n. (n is user input)
22. Write a program to display all natural numbers from 1 to n in reverse order.
23. Write a program to display all even numbers from 1 to n.
24. Write a program to display all odd numbers from 1 to n.

25. Write a program to find the sum of all natural numbers from 1 to n.
26. Write a program to find the sum of all even numbers from 1 to n.
27. Write a program to find the sum of all odd numbers from 1 to n.
28. Write a program to find the product of all natural numbers from 1 to n (factorial of n).
29. Write a program to display the multiplication table of a number n.
30. Write a program to display all multiples of a number m up to n terms.
31. Write a program to count how many numbers from 1 to n are divisible by 3.
32. Write a program to display all numbers from 1 to n that are divisible by 3 or 5.

Level 4 — Working with the digits of a number

33. Write a program to count the number of digits in a number n.
34. Write a program to display all the digits of a number n (one per line).
35. Write a program to find the sum of all digits of a number n.
36. Write a program to find the product of all digits of a number n.
37. Write a program to reverse a number n.
38. Write a program to find the largest digit in a number n.
39. Write a program to find the smallest digit in a number n.
40. Write a program to count the number of even digits and odd digits in a number n.
41. Write a program to check whether a number n is a palindrome (reads the same reversed).
42. Write a program to replace all zeros in a number n with the digit 5.
43. Write a program to find the sum of the first and last digit of a number n.

Level 5 — Number classification (prime, special numbers)

44. Write a program to read a number and check whether it is prime or not.
45. Write a program to display all prime numbers from 1 to n.
46. Write a program to display the first n prime numbers.
47. Write a program to check whether a number is an Armstrong number.
48. Write a program to display all Armstrong numbers from 1 to n.
49. Write a program to check whether a number is a perfect number.
50. Write a program to check whether a number is a strong number (sum of factorials of its digits).

51. Write a program to check whether a number is an automorphic number.
52. Write a program to check whether a number is a Harshad (Niven) number.
53. Write a program to find all factors (divisors) of a number n.
54. Write a program to count the number of factors of a number n.
55. Write a program to find the GCD (HCF) of two numbers.
56. Write a program to find the LCM of two numbers.

Level 6 — Series & patterns

57. Write a program to display the first n terms of the Fibonacci series.
58. Write a program to find the sum of the first n terms of the Fibonacci series.
59. Write a program to find the sum of the series 1 + 2 + 3 + ... + n.
60. Write a program to find the sum of the series 1^2 + 2^2 + 3^2 + ... + n^2.
61. Write a program to find the sum of the series 1 + 1/2 + 1/3 + ... + 1/n.
62. Write a program to find the value of x raised to the power y without using inbuilt power.
63. Write a program to print a right-angled triangle pattern of stars of height n.
64. Write a program to print an inverted right-angled triangle pattern of stars of height n.
65. Write a program to print a pyramid pattern of stars of height n.
66. Write a program to print a number triangle (row i contains numbers 1 to i).
67. Write a program to print Pascal's triangle for n rows.

Level 7 — Strings

68. Write a program to find the length of a string without using an inbuilt function.
69. Write a program to count the number of vowels and consonants in a string.
70. Write a program to count the number of words in a sentence.
71. Write a program to reverse a string.
72. Write a program to check whether a string is a palindrome.
73. Write a program to convert a string to uppercase and lowercase without inbuilt case functions.
74. Write a program to count the frequency of each character in a string.
75. Write a program to remove all spaces from a string.
76. Write a program to check whether two strings are anagrams of each other.

77. Write a program to find the first non-repeating character in a string.

78. Write a program to replace all occurrences of a character with another character in a string.

79. Write a program to toggle the case of each character in a string.

Level 8 — Arrays / Lists

80. Write a program to read n elements into an array and print them.

81. Write a program to find the sum and average of all elements in an array.

82. Write a program to find the largest and smallest element in an array.

83. Write a program to count the number of even and odd elements in an array.

84. Write a program to search for an element in an array (linear search).

85. Write a program to reverse the elements of an array.

86. Write a program to find the second largest element in an array.

87. Write a program to count the frequency of each element in an array.

88. Write a program to remove duplicate elements from an array.

89. Write a program to sort an array in ascending order (bubble sort).

90. Write a program to merge two arrays into one.

91. Write a program to find the sum of all even-indexed and odd-indexed elements separately.

Level 9 — Challenge round (mix everything)

92. Write a program to check whether a number is prime, using a function/method.

93. Write a program to print all prime numbers between two given numbers a and b.

94. Write a program to find the sum of digits of a number repeatedly until a single digit remains.

95. Write a program to count the number of prime digits present in a number n.

96. Write a program to check whether a number is a palindrome and a prime at the same time.

97. Write a program to find the largest and smallest number that can be formed using the digits of n.

98. Write a program to convert a decimal number into its binary equivalent.

99. Write a program to convert a binary number into its decimal equivalent.

100. Write a program to display a menu that lets the user repeatedly choose any of the above tasks until they choose to exit.

You made it to 100. Now you're ready for DSA. — codeWithShivah